Introduction to the Fir cluster

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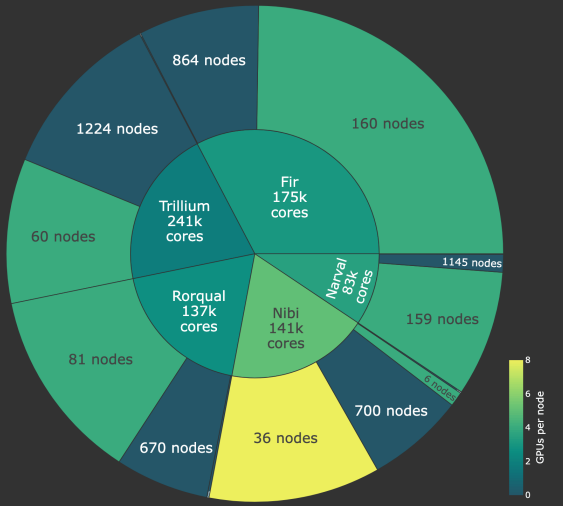
Intro

- Fir is hosted and operated by Simon Fraser University within Canada's national digital research infrastructure
- Replaces older Cedar cluster
- Fir was open to all users on August 12th
- Named after the Douglas fir, a conifer renowned for its towering height, symbolizing the cluster's computational power
- Compared to Cedar: 3-4X increase in CPU-core compute and memory, GPU number down but power up by 3.5X
  - in my tests, a single core is $\sim 5.5X$ faster than on Cedar
  - ranked #78 on the June 2025 TOP500 list $\leftarrow$ using a subset of the cluster during assembly
- Fir cluster is ready to support RAC'2025 allocations and general research workloads
- Some services are not set up yet – more on these later

Alliance's national HPC clusters

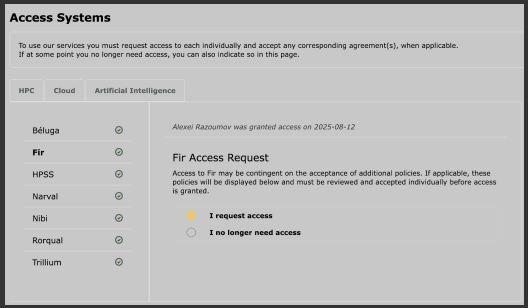
- 2025 infrastructure renewal `https://docs.alliancecan.ca/wiki/Infrastructure_renewal`
  - Cedar → Fir • Béluga → Rorqual
  - Graham → Nibi • Niagara → Trillium
  - Narval stays the same for now
- Total: 5 175 nodes, 776 912 cores, 2 152 GPUs
- Area size ∝ max theoretical single-precision FLOPS
  - plot shows overall max theoretical compute capability
  - in reality not so GPU-heavy ⇐ less than 100% GPU efficiency for complex problems + problem-dependent
  - not accurate for older CPUs/GPUs
  - plot does not include AI / other specialized clusters
- Partitions coloured by GPUs/node
- One Alliance account to access all systems
- Fir, Nibi, Rorqual, Narval are general-purpose clusters
- Trillium is for large parallel jobs: scheduling by node only, in multiples of 192 cores
- Most GPUs on the new clusters are NVIDIA's H100s
- Mostly identical software environment (via modules)
- Job sizes, max runtimes, filesystem specifications, quotas, and policies vary across clusters



- Interactive version of this plot
  `https://wgpages.netlify.app/files/clusters.html`

Access to Fir

- Similarly to the other national systems, users must request access via `https://ccdb.alliancecan.ca/me/access_systems`
  - i.e. log in to CCDB and then select Resources | Access Systems
  - if you have used Cedar recently, we have preemptively granted you access to Fir
  - can always verify your access at the address above
  - for certain systems, might need to accept agreements on the same page
  - can take up to 1hr for your access to be enabled



- Like you did for Cedar, connect via ssh to `fir.alliancecan.ca`
- 3 login nodes at the moment, randomly land on one of them
  - limits on user processes: similar to Cedar

System highlights

Details at `https://docs.alliancecan.ca/wiki/Fir`

- 175,104 CPU cores
- 864 base nodes: 192 cores, 750 GB memory, 2x AMD EPYC 9655 (Zen 5) processors @2.7 GHz
  - familiar ratio ~4GB/core for base nodes
  - best practices later in scheduling
- 8 large-memory nodes: 192 cores, 6000 GB memory, 2x AMD EPYC 9654 (Zen 5) processors @2.4 GHz
- 160 GPU nodes: 48 cores, 1125 GB memory, 1 x AMD EPYC 9454 (Zen 4) processor @2.75 GHz
- 51 PB high-performance DDN Lustre storage with 3 partitions: `/home`, `/scratch`, `/project`
- InfiniBand NDR interconnect
- Operating System: AlmaLinux 9.6
  - familiarity with the Linux command line is essential $\Rightarrow$ attend our courses (Intro to bash command line + Intro to HPC) throughout the year
- Direct-to-chip liquid cooling
- Full access to the internet from the compute nodes

Filesystems

Details in the table at `https://docs.alliancecan.ca/wiki/Fir#Storage`

- 51 PB high-performance DDN Lustre storage: no spinning drives now, all SSD-based
- Shared by 3 mounts (`/home` , `/scratch` , `/project`), with different quotas and policies
- No automatic compression and deduplication
  - built-in compression on physically separate `/nearline` ← separate slide
- Where to find `/scratch` and `/project`) directories
  - `$SCRATCH` and `~/scratch` both point to `/scratch/$USER`
  - `$PROJECT` points to `~/project` that links to your default project `/project/<gid>`
  - `~/projects` contains links to all your projects `/project/{<gid1>,<gid2>,...}`
- `/scratch` purge policy: same as on Cedar, following the Alliance's procedure
  - ∼6 weeks before purging files, several email warnings
- Every group has a `/project` directory with a 1 TB quota
  - `/project` filesystem uses the new per-project quota, rather than file's GID

  ⇒ all files inside of the project apply against the PI's quota independently of the UID/GID

  ⇒ this eliminates issues often faced with tools like `git` , `mv` , or `tar`
  - ideally, request an increase via RAC
  - can temporarily increase your regular (non-RAC) allocation

Quotas and data transfer

- Command `quota [--home] [--scratch] [--project] [--nearline]`
  - shorter alias for `diskusage_report`
- Cedar $\longrightarrow$ Fir data migration already happened
  - no action required, as the two clusters shared the same mount points for a while
- For `rsync` / `scp` transfer, use the login node $\leftarrow$ no dedicated data transfer node(s) for now
- Globus endpoint: `alliancecan#fir-globus`

Local scratch

- Now 7TB SSD on each compute node
- Inside jobs `$SLURM_TMPDIR` points to `/localscratch/${USER}.${SLURM_JOBID}.0`
- Must copy data out before your job ends
- On both login and compute nodes, `/tmp` is a RAM disk for short-term usage
  - any file(s) written to `/tmp` will count toward cgroup's memory usage

Tape-based nearline

- Cedar's Nearline should be automatically available on Fir
  - moved from Cedar as is, no new hardware
  - by default available to PIs only, default quota ~10TB (policy still under discussion)
- Use `tar` or `dar` to create an archive directly in `/nearline`, keep the source files in their original location elsewhere; ideal file size: 10GB - 4TB

```
NEAR=~/nearline/def-razoumov-ac/razoumov
cd ~/projects/def-razoumov-ac/razoumov/visThis2019-airfoil
dar -s 10G -w -c $NEAR/monday $(for d in processor?; do echo "-g␣$d"; done)   # two slices: 10G and 8.2G
dar -l $NEAR/monday > $NEAR/monday.index   # create an index file; can be stored in /nearline
```

- No need to compress your data (built-in hardware compression), Ok with already compressed files
- Only available on login and data-transfer (preferred, coming later) nodes, not on compute nodes
- After ~1d, the system will copy the file(s) to tape

```
quota --nearline --per_user --all_users   # check the Location field, e.g. ''On disk and tape''
```

- Later the disk copy may be deleted, although the file(s) will still appear in the the directory listing

```
lfs hsm_state $NEAR/monday.?.dar   # to check file's location, e.g. ''exists archived''
```

- To mark a file for restoring, either run `lfs hsm_restore <file>` or simply try to access it
- Might take over 1h to retrieve from tape
- Tape copy of deleted files will be retained for up to 60d, but will need the file's full path to restore

Networking

- InfiniBand NDR (Next Data Rate) interconnect
- Base 864 nodes grouped into 4 islands
- Between the islands:
  - 27:5 blocking factor, i.e. will saturate at 5/27 of max theoretical speed when using network simultaneously from all nodes
- Inside each island
  - at least 100 Gb/s and symmetric for all nodes in pairs, ∼90 ns port-to-port latency
  - 216 nodes per island organized into 108 trays (2 nodes per tray)
  - single 200 Gb/s to the first node in each tray with another followup cable to the second node
- GPU nodes utilize 2:1 blocking factor
- Storage access is fully non-blocking

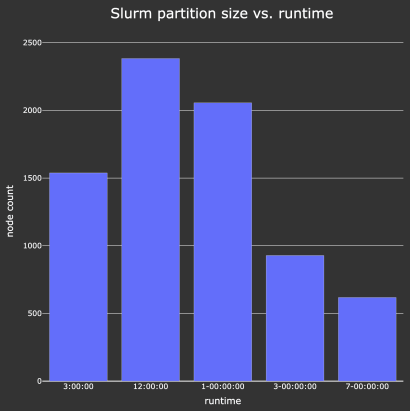
Software

- Mostly the same as on Cedar
- StdEnv/2023 is now default (StdEnv/2020 still there but hidden)
- cuda/12.6 is default on all new national clusters
- Recompile your code
- Redo the scalability test of your CPU parallel code
- Reinstall your Python virtual environments
- In my case, on first login it was complaining about loading an old intel/2020.1.217
  $\Rightarrow$ had to run:

```
module reset   # restore modules from your "default" or system's default
```

Scheduler

- No new Slurm features compared to Cedar
- Jobs runtimes now range from 1hr to 7d
  - >7d jobs no longer allowed across the Federation
  - highly unlikely that we'll enable 28d jobs like in the past
- No CPU overcommit in Slurm (now or in the plans)



- processing output of `sinfo`
- 43 Slurm partitions at the moment, grouped here into 5 "runtime bins"
- many nodes can accept jobs from multiple partitions, so the total number of nodes in the plot exceeds the physical number of nodes

864 base CPU nodes: memory architecture and scheduling

- Each node: 192 cores, 750 GB memory, 2x AMD EPYC 9655 (Zen 5) processors @2.7 GHz
- Two sockets with 96 CPU cores each, organized into 12 chiplets
- Each chiplet is a separate NUMA (non-uniform memory access) node with 8 CPU cores
- Memory attached to a socket $\Rightarrow$ all chiplets/cores within the same socket have uniform local memory access (subject to local bandwidth limitations), slower remote memory access to the other socket
  - in other words, memory latency/bandwidth depend on core and memory locality
- `#SBATCH --cpus-per-task=8` will ensure that all threads in a task stay within a single NUMA node $\longleftarrow$ ideal for memory-bound tasks
- `#SBATCH --ntasks-per-node=24` will launch 24 tasks per node, to fully utilize all chiplets on a node when combined with the previous flag (assuming your code can scale to 192 cores)

160 GPU nodes: memory architecture and scheduling

- Each node: 48 cores, 1125 GB memory, 1 x AMD EPYC 9454 (Zen 4) processor @2.75 GHz
- One socket with 48 cores, organized into 6 chiplets and 4 NUMA nodes, i.e. NUMA nodes don't align perfectly with chiplets

  $\Rightarrow$ each chiplet is 8 cores, and each NUMA node is 1.5 chiplets (12 CPU cores)
- If your job is memory-bound, you want all your threads in a task to stay within a single NUMA node
  - `#SBATCH --cpus-per-task=12` to bind tasks to NUMA nodes
  - combine with `#SBATCH --ntasks-per-node=4` if need better utilization of the entire node
- `#SBATCH --cpus-per-task=8` might be even better if compatible with your code / workflow
  - will place all threads within one chiplet and within one NUMA node
  - final recommendation depends on each code and how much data your send to/from GPU

    $\Rightarrow$ study your code's performance with both
- 4 NVidia H100 80GB accelerators per each GPU node $\Rightarrow$ 640 GPUs total

GPUs: full cards

- Cedar $\longrightarrow$ Fir transition from older P100 / V100 / A100 to fewer modern H100 GPUs
- Four H100 GPUs per node
- Unlike on Cedar, on Fir *must* specify the GPU type
- Use either `--gpus-per-node=<gpuType>:<N>` or `--gpus=<gpuType>:<N>`
- Currently four "GPU types" are supported:
  - `h100` $\leftarrow$ full H100-80gb
  - `nvidia_h100_80gb_hbm3_3g.40gb` $\leftarrow$ MIG partition
  - `nvidia_h100_80gb_hbm3_2g.20gb` $\leftarrow$ MIG partition
  - `nvidia_h100_80gb_hbm3_1g.10gb` $\leftarrow$ MIG partition
- For jobs that can utilize full GPUs, use one of these:
  - `--gpus-per-node=h100:<N>` to request N full H100-80gb GPUs per node, where N=1..4
  - `--gpus=h100:<N>` to request N full H100 GPUs spread anywhere on the cluster
  - usage billed at 12.2 RGUs (Reference GPU Units)
- Vast majority of codes won't be able to fully utilize an H100 card ... which is a problem for a $\sim$USD25k card

Why a job might not utilize an H100 GPU efficiently?

- The computational problem is too small to saturate the GPU
- Bursty GPU usage, e.g. due to many serial parts in the code, or too much compute on the host (CPU-bound) for data pre-processing
- Overhead from too many tiny kernel launches
- The computational problem is inherently memory-bound instead of compute-bound
- Other data transfer issues, e.g.
  - NUMA locality (CPU cores on a different socket than the GPU)
  - NVLink bandwidth
  - slow input data read from network storage
- Using old/unoptimized libraries (lack of optimization for H100)
- Other constraints / bottlenecks resulting in an intermittent GPU utilization pattern

Estimating GPU utilization

- A (GPU) job monitoring portal on Fir is coming, but not earlier than the end of September
  - for now can use the portals on Nibi (`https://portal.nibi.sharcnet.ca`) or Rorqual (`https://metrix.rorqual.alliancecan.ca`), as these clusters have exactly the same H100 cards
- User access to GPU performance counters on Fir is currently disabled
  ⇒ GPU profiling by user directly on the system is not available
  - multi-tenant security reasons: could expose things like memory access patterns, instruction throughput
  - same on Nibi
  - on Narval and Rorqual, GPU profiling is possible but requires disabling DCGM (NVIDIA's Data Center GPU Manager to collect metrics) during job submission
  - details in `https://docs.alliancecan.ca/wiki/Using_GPUs_with_Slurm#Profiling_GPU_tasks`
- At the moment, the only option on Fir is to attach to an active GPU job to run `nvidia-smi`
  - `nvidia-smi` does not require GPU performance counters to report GPU utilization, takes it directly from the driver's activity monitoring
  - currently shows GPU utilization on full H100 cards, along with the total power usage (out of max 700W)
  - GPU utilization on MIGs should be enabled later
  - bash functions to simplify running this from the login node:

```
function job-nvidia-smi() {
  srun --jobid=$(squeue -u $USER | awk 'NR==2{print$1}') --pty nvidia-smi
}
function job-nvidia-smi-continuous() {
  srun --jobid=$(squeue -u $USER | awk 'NR==2{print$1}') --pty watch -n 1 nvidia-smi
}
```

GPU utilization via `nvidia-smi` on Fir

```
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.65.06              Driver Version: 580.65.06      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA H100 80GB HBM3          On  |   00000000:4C:00.0 Off |                    0 |
| N/A   34C    P0            268W /  700W |   76827MiB /  81559MiB |    100%      Default |
|                                         |                        |             Disabled |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|    0   N/A  N/A         4078424      C   ...eFactorization/primesGPU_real      76816MiB |
+-----------------------------------------------------------------------------------------+
```

GPU utilization via the job monitoring portal on Nibi

1. Run your code on Nibi

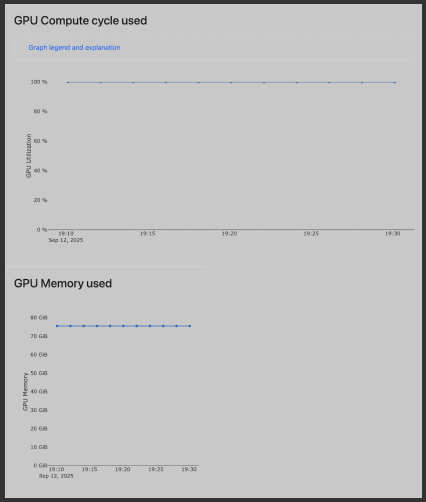
```
source ~/startSingleLocaleGPU.sh    # normally should be
                                    # 'module load cuda' + your code
cd ~/chapelCourse/primeFactorization
chpl --fast primesGPU.chpl
bat gpu.sh
```

```
-------------------------------------------------------------
       | File: gpu.sh
-------------------------------------------------------------
   1   | #!/bin/bash
   2   | #SBATCH --time=00:60:00
   3   | #SBATCH --mem-per-cpu=3600
   4   | #SBATCH --gpus-per-node=h100:1    # full H100
   5   | #SBATCH --account=def-someuser
   6   | ./primesGPU --n=10_000_000_000

   sbatch gpu.sh
```

   - caution: run for at least 10m, as data points are only collected occasionally, and you'll need enough of them for a *reliable graph*

2. Log in to `https://portal.nibi.sharcnet.ca`
3. Job stats, select your jobID, scroll down to GPU section



GPU utilization via Nsight package profiler on Rorqual

NVIDIA's Nsight GUI profiler

1. Generate a profiling report on the GPU

```
source ~/startSingleLocaleGPU.sh   # normally should be 'module load cuda' + your code
cd ~/chapelCourse/primeFactorization
chpl --fast -g --gpu-ptxas-enforce-optimization primesGPU.chpl
DISABLE_DCGM=1 salloc --time=0:60:0 --mem-per-cpu=3600 \
               --gpus-per-node=h100:1 --account=def-someuser
while [ ! -z "$(dcgmi␣-v␣|␣grep␣'Hostengine␣build␣info:')" ]; do  sleep 5; done
                                                    # wait until DCGM is disabled
nsys profile --trace=cuda -o 111 ./primesGPU -nl 1 --n=10_000_000_000
>>> exit the interactive job after the calculation is done
```

2. Use GUI profiler via `https://jupyterhub.rorqual.alliancecan.ca`
   - log in, select JupyterLab interface, no GPU needed
   - launches a Slurm job – wait for it to start and then for the JL server to start
   - start a desktop, open a terminal, type the commands below

```
module load cuda/12.6
cd ~/chapelCourse/primeFactorization
nsys-ui 111.nsys-rep
```

If your code's GPU utilization is $\lesssim 75\%$, what can you do?

1. scale up your problem to fill the GPU, keeping an eye on the utilization number – might be easier said than done
2. use a MIG partition (next slide)
3. compute on CPU cores

GPUs: static MIG partitions

- If your job uses $< \frac{1}{2}$ compute and $< \frac{1}{2}$ available memory of a full GPU $\Rightarrow$ run it on a MIG partition

> ∼half of all GPUs on Fir are partitioned with MIG (Multi-Instance GPU) technology into 8 smaller virtual GPU instances of which 7 are available to users $\Rightarrow$ replace the GPU flag in your script with one of these:

- `--gpus=nvidia_h100_80gb_hbm3_1g.10gb:1` to request one 1g.10gb partition, at 1.74 RGUs
- `--gpus=nvidia_h100_80gb_hbm3_2g.20gb:1` to request one 2g.20gb partition, at 3.48 RGUs
- `--gpus=nvidia_h100_80gb_hbm3_3g.40gb:1` to request one 3g.40gb partition, at 6.1 RGUs
- Usage billed accordingly, resulting in higher priority compared to allocating a full GPU
- Do not ask for multiple MIG instances in a single job
  1. on the same GPU: not supported in our Slurm config $\Rightarrow$ to run multiple independent tasks, use MPS (next slide) rather than MIG
  2. across multiple GPUs: no support in MIG for data transfer between GPUs over NVLink / NVSwitch
- No graphics APIs (OpenGL, Vulkan) are supported in MIGs
- Our wiki mentions the max number of CPU cores per GPU type: 1 for H100-1g.10gb, 3 for H100-2g.20gb, 6 for H100-3g.40gb, 12 for H100-80gb
  - currently not enforced, but a good practice to follow, otherwise you'll be billed for higher usage
  - may be enforced in the future
- MIG details at `https://docs.alliancecan.ca/wiki/Multi-Instance_GPU`

GPUs: dynamic sharing via MPS (Multi-Process Service)

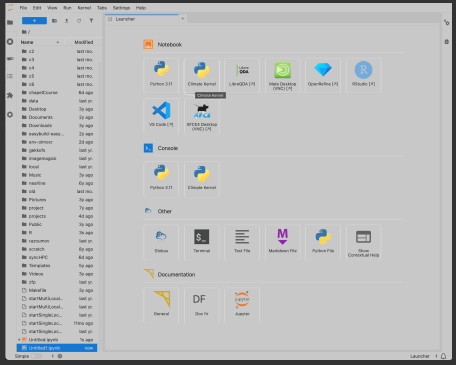
- Alternative technology: dynamic sharing of a GPU (cores + memory) between unrelated processes
  - all processes must belong to the *same user*: a GPU can't be safely shared with MPS between different users
- No need to rewrite or even recompile your code!
- Any number of MPS clients (processes) between 2 and 48 (for A100 and H100 GPUs)
  - much higher granularity than MIG
  - e.g. a group of MPI ranks sharing a single GPU
  - or multiple serial GPU codes (GPU farming)
- Launch the MPS daemon from inside your Slurm job submission script
  - all user processes (including the daemon) are killed automatically at the end of the job
- MPS can be combined with MIGs
- Details at `https://docs.alliancecan.ca/wiki/Hyper-Q_/_MPS`

Visualization

- H100 GPUs do not support graphics rendering efficiently ⇒ don't use them for visualization
  - OpenGL / Vulkan pipelines can run only on 2/66 thread controllers on an H100 ⇒ rendering will utilize the card only at ∼ 3% efficiency ... not the best use of this expensive device
- Instead, we recommend using CPU nodes for visualization in the vast majority of cases
  - much faster to load *partitioned* data in parallel onto 16/32/64/128 CPU cores ⇒ why not use the same cores for parallel rendering?
  - far more CPU cores than GPUs ⇒ much easier to allocate for batch rendering
  - modern libraries can do both rasterization and ray tracing on CPU cores
  - we teach remote parallel visualization (interactive + batch off-screen) in our courses
- Nibi has AMD MI300A nodes with unified CPU+GPU memory – to be tested for vis
- SHARCNET is planning to make some older T4's available for rendering (interactive jobs or Open OnDemand) on Nibi

Interactive access

- ~ 10 nodes in the interactive partitions:
  - `sinfo | grep interac`
  - 3hr max runtime
  - use `salloc` to start interactive jobs
- `https://jupyterhub.fir.alliancecan.ca` now online
  - log in, select JupyterLab interface and other parameters
  - launches a Slurm job – wait for it to start and then for the JL server to start
- Open OnDemand (OOD) is planned, but no firm details yet on what services it'll offer
  - will use a number of dedicated nodes outside of Slurm
  - might or might not add a Slurm partition to it later
  - lower priority; end of October?



- Can also use graphical interfaces on other clusters:
  - `https://jupyterhub.rorqual.alliancecan.ca` will start a Slurm job
  - `https://ondemand.sharcnet.ca` on Nibi
  - `https://ondemand.scinet.utoronto.ca/pun/sys/dashboard` on Trillium

Fir Cloud

- The team is working on it, planning to make it available towards the end of September
  - there will be a separate email update from the Alliance when it is ready
  - watch `https://docs.alliancecan.ca/wiki/Cloud#Cloud_systems`
- For now RAC'2025 cloud allocations remain on Cedar Cloud `https://cedar.cloud.computecanada.ca`, until Fir Cloud is open
- Will provide RAC + default allocations (same as Cedar cloud)
  - default allocations will not enabled by default $\Rightarrow$ submit a ticket support@tech.alliancecan.ca
- Fir Cloud and Arbutus should be very similar, more so than Cedar and Arbutus

Summary

- Fir is ready for production, both in general queues and RAC allocations
- Minimal changes in software and scheduling, use Fir as you would use Cedar
- May need to recompile your code
- Pay attention to parallel scalability of your CPU code
- Biggest changes: GPU scheduling and utilization, MIG partitions
- Some features still in the pipeline: job monitoring portal, Fir Cloud, Open OnDemand, dedicated data transfer node(s)
- Cedar should be fully decommissioned as of Sep-12
  - with the exception of Cedar Cloud for now
  - part of the old cluster will be repurposed as an internal SFU resource for researchers and students
- Watch Sergey Mashchenko's (SHARCNET) July 2025 webinar *Migrating to the upgraded national systems*
- Any problems ⇒ email support@tech.alliancecan.ca to open a ticket
  - as usual, provide as much detail as you can: specific error messages, jobID, Slurm script, software version, etc.

Questions?